

assignment #02

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
Professor: Dr. Michael Caracotsios
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Student names: _____

Due Date: Friday October 10th, my Mailbox (3:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name(s) in the report.
 - Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
 - Bring a hard copy of your report as a PDF file by the Due Date and Time in class
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- Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Optimal Cooling Water Flow Rate in Condenser

Your engineering team has been asked to design a single-pass condenser for the condensation of propane vapor at 20 bar pressure. The vapor is at its dew point, and the flow rate is 2088 kmol/hr. Cooling water at 25 °C is available from a nearby cooling tower. Estimate the optimal water circulation rate in kg/s.

Data:

Heat Exchanger Type: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger (\$ January 2010): $C_{PURCHASE} = 28,000 + 54.0 A^{1.2}$ $A [=] m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost (\$/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$14.8 / 1000 m^3$

Overall Heat Transfer Coefficient: $U = 425.6 W / (m^2 \cdot K)$

Heat Capacity of Water: $C_p = 4.184 kJ / (kg \cdot K)$

Operating Year: 8000 hrs

Cost Escalation Factor: 1.50

Problem B: Optimization of a refrigeration loop

Methane at temperature 25 °C and pressure 30 bar is to be precooled down to -38 °C prior to liquefaction. The precooling will be done using propane as the refrigerant. Design a multi-stage refrigeration loop and calculate the annual operating cost and the capital cost. You may attempt to minimize the total annualized cost for your design. The flow rate of methane is 250 MMSCFD. The isentropic efficiency of the compressors can be taken as 80% and the mechanical efficiency equal to 95%.

Notes:

For both problems assume that the physical properties of methane and propane are best described by the Peng-Robinson Equation of State

Minimum temperature approach for condenser may be taken as $\Delta T_{\min} = 10^\circ\text{C}$
Minimum temperature approach for evaporators:

Stage	$\Delta T_{\min}$
1	8
2	6
3	4
4	2

Gavin Towler and Ray Sinnott, **Chemical Engineering Design**. Principles Practice and Economics of Plant and Process Design, Second Edition © Elsevier Ltd. 2013.

Data:

Heat Exchangers: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger (\$ January 2010): $C_{\text{PURCHASE}} = 28,000 + 54 A^{1.2}$ $A [=] m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Compressors: Centrifugal

Purchase cost of the Compressor (\$ January 2010): $C_{\text{PURCHASE}} = 580,000 + 20,000 P^{0.6}$ $P [=] kW$

Power Requirement: P

Lang Factor: $LF = 2.5$

Valves: Expansion Valves

Purchase cost of the Valve (\$ January 2010): $C_{\text{PURCHASE}} = 8,142 m^{0.7}$ $m [=] kg / s$

Mass Flow Rate: m

Lang Factor: $LF = 1.3$

Drums: Flash Drum with Hemispherical Head and Bottom

Purchase cost of the Drum (\$ January 2010): $C_{PURCHASE} = 11,600 + 34W^{0.85}$ $W [=] kg$

Shell Mass: W

Lang Factor: $LF = 4.0$

Data (Continued):

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost (\$/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$0.354 / GJ$

Cost of Electricity: $C_{EL} = \$0.06 / kWh$

Overall Heat Transfer Coefficient Condenser: $U = 425.6 W / (m^2 \cdot K)$

Overall Heat Transfer Coefficient Evaporators: $U = 425.6 W / (m^2 \cdot K)$

Heat Capacity of Water: $C_p = 4.184 kJ / (kg \cdot K)$

Cooling Water Temperature: $T_{CWin} = 25 ^\circ C$

Cooling Water Pressure: $P_{CWin} = 20 psia$

Cooling Water Temperature Rise: $10 ^\circ C$

Operating Year: $8000 hrs$

Cost Escalation Factor: 1.50

Additional Instructions for your Report:

Your Custom table should contain three tabs

Tab 1 (Name KPIs): Display COP, Total Power(kW), Total Annualized Capital Cost (MM\$/oper_year), Total Utility Cost (MM\$/oper_year) and the sum of the two costs

Tab2 (Name: Detailed Cost): The installed cost (\$) of each equipment and cost of each utility in \$/oper_year

Tab3 (Name: Optimal Pressure): The final discharge pressure of each stage in bar